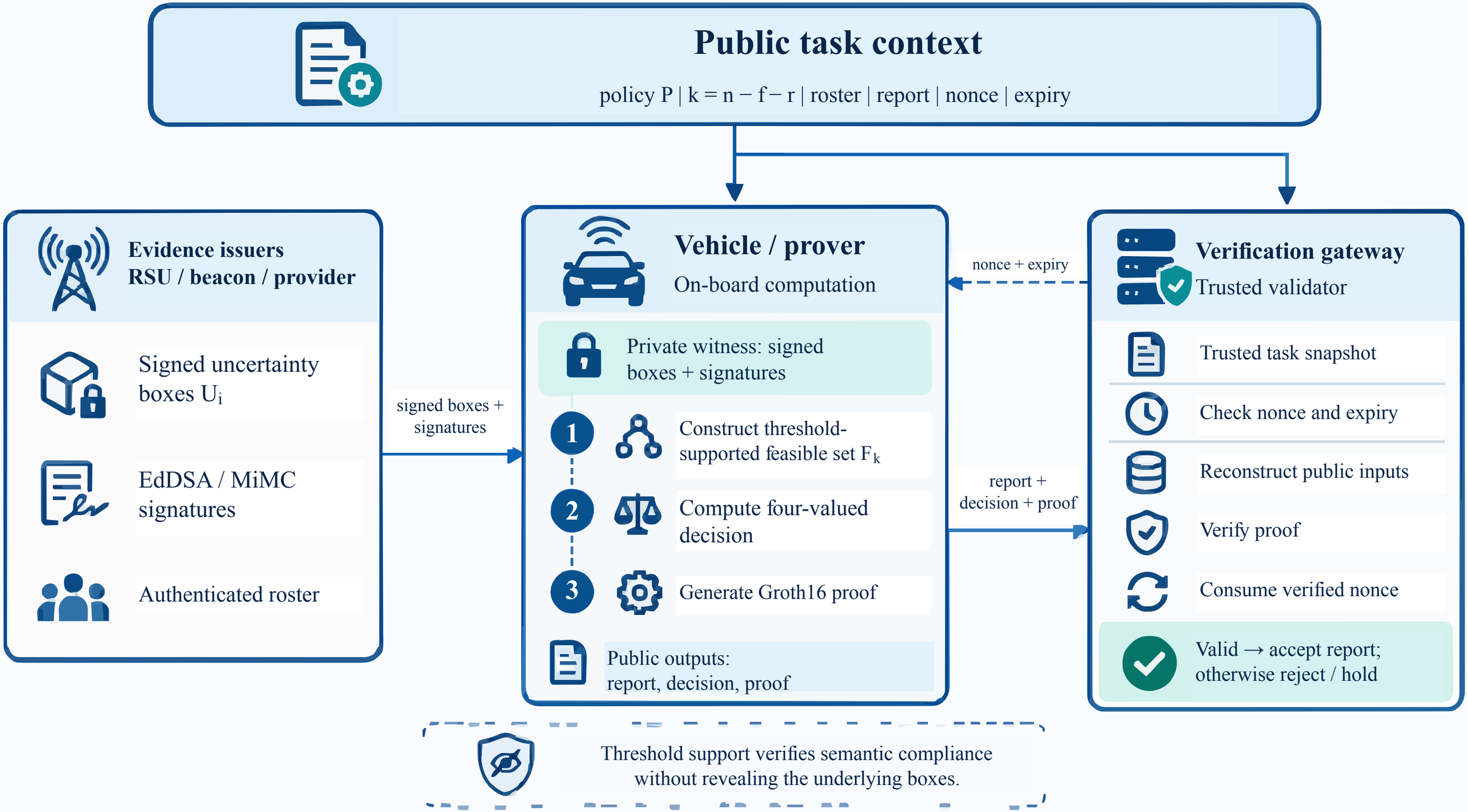

Public task context
policy P | k = n − f − r | roster | report | nonce | expiry
Evidence issuers
RSU / beacon / provider
Signed uncertainty boxes U_i
EdDSA / MiMC signatures
Authenticated roster
signed boxes + signatures
Vehicle / prover
On-board computation
Private witness: signed boxes + signatures
1
Construct threshold-supported feasible set F_k
2
Compute four-valued decision
3
Generate Groth16 proof
Public outputs: report, decision, proof
nonce + expiry
report + decision + proof
Verification gateway
Trusted validator
Trusted task snapshot
Check nonce and expiry
Reconstruct public inputs
Verify proof
Consume verified nonce
Valid → accept report; otherwise reject / hold
Threshold support verifies semantic compliance without revealing the underlying boxes.